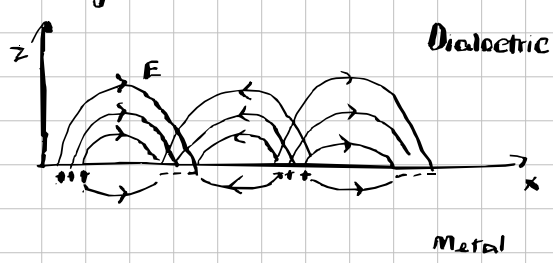
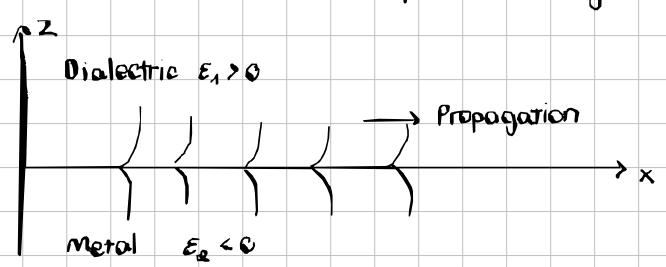
Plasmonics

What is surface plasmon? Think about the electrons inside a metal.
Inside a dielectric, electrons are generally bound to atoms. In a metal, conduction electrons can move collectively in response to an external electric field.
Imagin electrons accumulate in one region of a metal surface, leaving another region relatively positive, the resulting electric force tends to restore the charge distribution
Sine electrons have inertia, they overshoot and oscillate. This is the origin of a collective electron oscillation called <u>plasmon</u>



When this collective charge oscillation is coupled to an electromagnetic field propagating along the surface we obtain a <u>surface plasmon polariton</u>. Polariton emphasizes that the mode combines electronic motion and an electromagnetic field.

- Along the surface, the wave travels.
- Perpendicular to the surface, its amplitude decays



Consider the fields

$$\vec{E}_j = \vec{E}_{0j} e^{i(\vec{k}_j \cdot \vec{r} - \omega t)} \qquad \vec{k}_1 = (k_x, 0, i\alpha_1) \qquad \vec{k}_2 = (k_x, 0, -i\alpha_2)$$

Above the surface $z > 0$

$$\vec{E}_1 = \vec{E}_{01} e^{i(k_x x - \omega t)} e^{-\alpha_1 z}$$

Below the surface $z < 0$

$$\vec{E}_2 = \vec{E}_{02} e^{i(k_x x - \omega t)} e^{\alpha_2 z}$$

The imaginary perpendicular wavenumber does not mean the wave stops propagating. It means the field is spatially confined in that direction
At the interface, Maxwell's boundary equations require

$$E_{1x} = E_{2x} \qquad H_{1y} = H_{2y} \qquad \varepsilon_1 E_{1z} = \varepsilon_2 E_{2z}$$

From Ampere's law for a plane wave

$$\frac{1}{\varepsilon_j} \nabla \times \vec{H} = \frac{1}{c} \frac{\partial}{\partial t} \vec{E} \qquad \Rightarrow E_{jx} = \frac{k_{jz}}{\omega \varepsilon_0 \varepsilon_j} H_{jy}$$

Because E and H are continuous

$$\frac{k_{1z}}{\varepsilon_1} = \frac{k_{2z}}{\varepsilon_2} \Rightarrow \frac{\varepsilon_1}{\varepsilon_2} = -\frac{\alpha_1}{\alpha_2}$$

Thus, a bound SPP at an interface requires $\varepsilon_2 \propto -\varepsilon_1$
Derive SPP disporsion, in each medium, we obtain

$$k_x^2 - \alpha_1^2 = \frac{\omega^2}{c^2} \varepsilon_1$$

$$k_x^2 - \alpha_2^2 = \frac{\omega^2}{c^2} \varepsilon_2$$

Together with $\frac{\varepsilon_1}{\varepsilon_2} = -\frac{\alpha_1}{\alpha_2}$

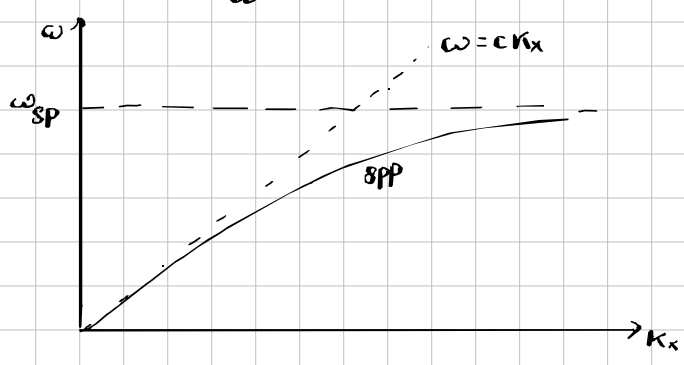
$$k_x = \frac{\omega}{c} \sqrt{\frac{\varepsilon_1 \varepsilon_2}{\varepsilon_1 + \varepsilon_2}}$$

In Drude Model,

$$\varepsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2}$$

For vacuum above the metal $\varepsilon_1 = 1$

$$\omega_{sp} = \frac{\omega_p}{\sqrt{2}}$$

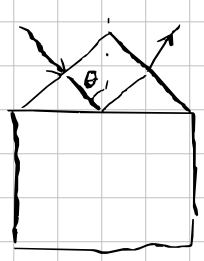


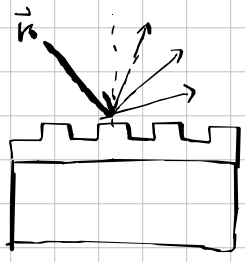
At small k_x: $\omega \approx ck_x$
This behaves as free space electromagnetic wave
At large k_x: $\omega \to \omega_{sp}$
Since wavelength is $\lambda_{\parallel} = 2\pi/k_x$. Surface wavelength is increasingly small
Since decay grows with k_x, E field becomes more highly concentrated around surface.

Ordinary light cannot excite SPP. At a given frequency, an SPP has a greater in-plane momentum than a free photon

$$k_{SPP} > \frac{\omega}{c} \sqrt{\varepsilon_1}$$

A photon incident from dialectric has $k_{\parallel} = \frac{\omega}{c} \sqrt{\varepsilon_1} \sin\theta$
This is momentum mismatch. To overcome this, we need prism coupling and Grating coupling





Planar SSP
Propagating surface mode
Travels along surface

Nanoparticle plasmon
Localized Resonance.
Electrons oscillate within particle

Real metals have damping so SPP can only propagate along the surface only a finite distance before its energy is dissipated

Previously

$$k_{spp} = \frac{\omega}{c}\sqrt{\frac{\varepsilon_1 \varepsilon_2}{\varepsilon_1 + \varepsilon_2}} \qquad \varepsilon_1 \text{ dialectric permittivity} \quad \varepsilon_2 \text{ metal permitivity}$$

For a lossless model

$$\varepsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2}$$

Conduction electrons also experience collisions and other relaxation processes

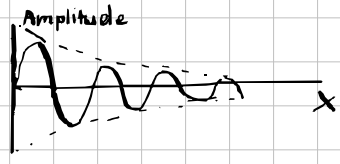
$$\varepsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega + i\gamma)}$$

ε_2 becomes complex : $\varepsilon_2 = \varepsilon_2' + i\varepsilon_2''$

Consequently, SPP wavenumber becomes complex $k_{spp} = \kappa' + i\kappa''$

The field on the surface travels as

$$E(x,t) = E_0 e^{i(k_{spp}x - \omega t)}$$

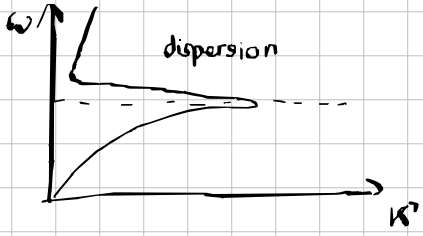


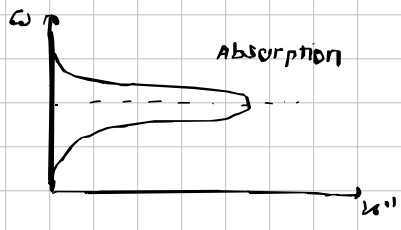
Substitute $k_{spp} = \kappa' + i\kappa''$

$$E(x,t) = E_0 e^{-\kappa'' x} e^{i(\kappa' x - \omega t)}$$

In the damped model

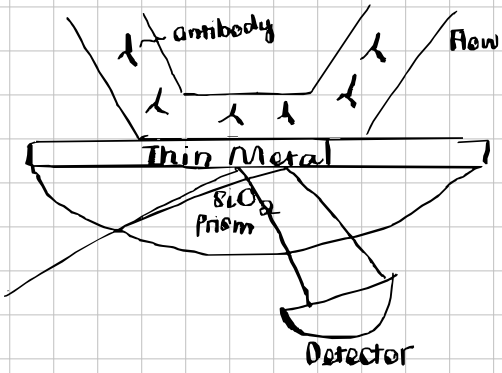
$$\varepsilon_1 + \varepsilon_2 \not\to 0 \text{ as } \kappa' \to \infty$$





SPP Application

When molecules bind to a metal surface, they change the local optical environment. This changes SPP dispersion relation, allowing us to detect molecular binding using light, without fluorescent labels



Prism allows light to excite an SPP
A thin metal film that supports spp
Antibodies immobalized on gold surface
A flow channel carrying antigens that may be bind to those antibodies.

Light enters the prism and excites an SPP at the metal-sample interface.
The SPP electromagnetic field extends a short distance into the sample
Antigens in the flow channel bind to antibodies on the gold surface.
This binding changes the refractive index experienced by the evanescent field.
Consequently, the SPP resonance condition changes

Since SPP wavenumber is $k_{spp} = \frac{\omega}{c}\sqrt{\frac{\varepsilon_m \varepsilon_d}{\varepsilon_m + \varepsilon_d}}$

The prism provides $k_{//} = \frac{\omega}{c} n_p \sin\theta$

At certain angle, light couples

After antigens bind, the local dielectric environment changes

$$\varepsilon_d \rightarrow \varepsilon_d + \Delta\varepsilon_d$$

when $\Delta\varepsilon_d \neq 0$, $\Delta k_{spp} \neq 0$

This means that even though the metal itself has not changed, supported mode has changed



And $\varepsilon_d \uparrow$ leads to $\omega_{sp} = \frac{\omega_p}{\sqrt{1 + \varepsilon_d}} \downarrow$

